

A Quick Guide to start Multi Agent Systems

Contributor: Ravi Kiran

This simulink tutorial gives you the working knowledge of Multi Agent Systems/ Connected systems.

Simple Linear MAS:

1. Consider an agent with following dynamics *moving on its own (Leader Agent)*, with acceleration input ' u_a '.

$$\begin{aligned}\dot{x}_1 &= u_a \\ \dot{x}_2 &= x_1\end{aligned}$$

Subsequently from the above dynamics, the x_1 becomes velocity and x_2 becomes the position of the vehicle.



2. Now there exists another agent (*follower*) moving according to the movement of the Leader. Its dynamics are given as:

$$\begin{aligned}\dot{y}_1 &= u_{f1} \\ \dot{y}_2 &= y_1\end{aligned}$$

Here u_{f1} is calculated based on the position and velocity of leader as:

Case1: To obtain consensus with Leader

$$u_{f1} = -K_1(y_1 - x_1) - K_2(y_2 - x_2).$$

Case2: To maintain a distance of 10m with Leader

$$u_{f1} = -K_1(y_1 - x_1) - K_2(10 + y_2 - x_2).$$

Where K1 and K2 are the proportional constants. Try changing the values of K1 and K2 and observe the effects. The value of 10 can be changed according to the distance requirement.

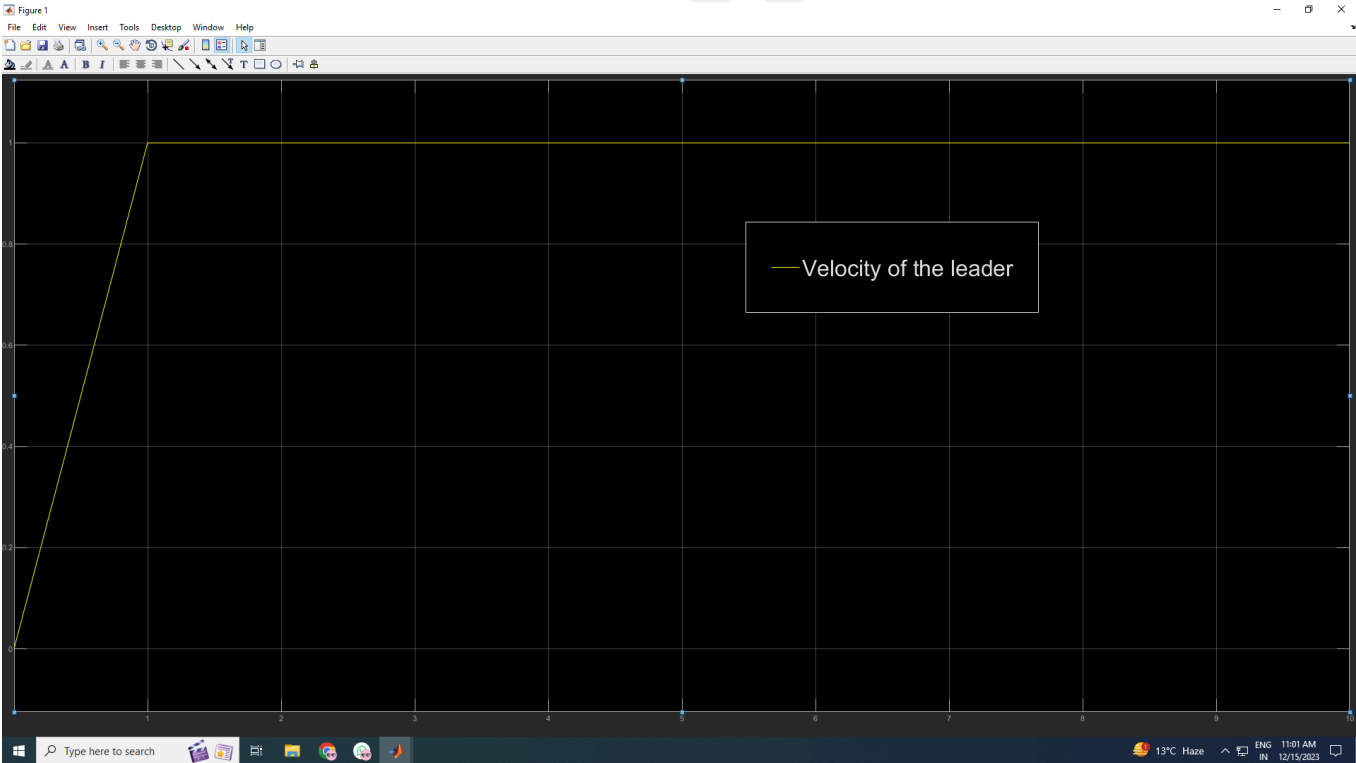
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3. Now let's start the simulation. Consider the leader agent initially at rest. Reaches a constant linear velocity of 1m/s in 1 second. It's acceleration (u_a) input and velocity can be given as:

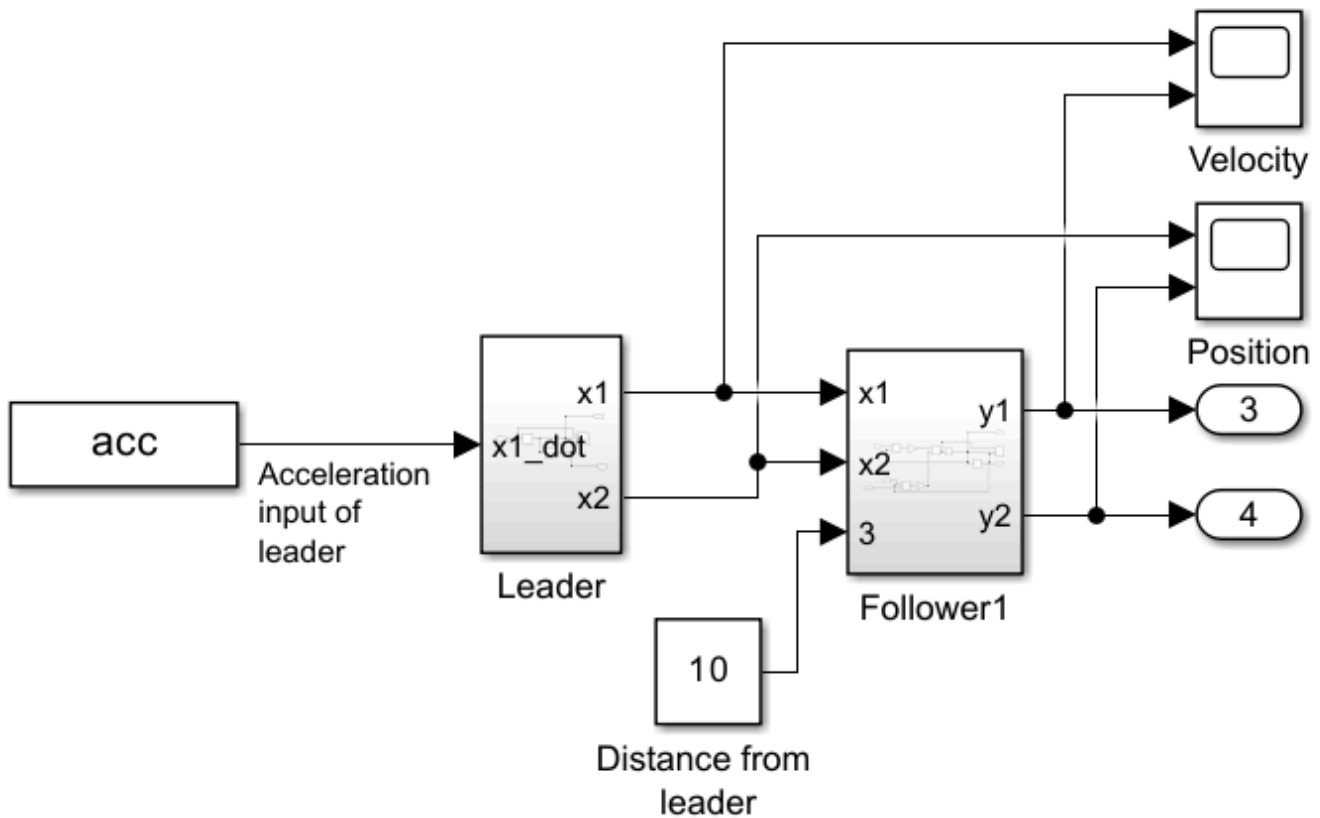
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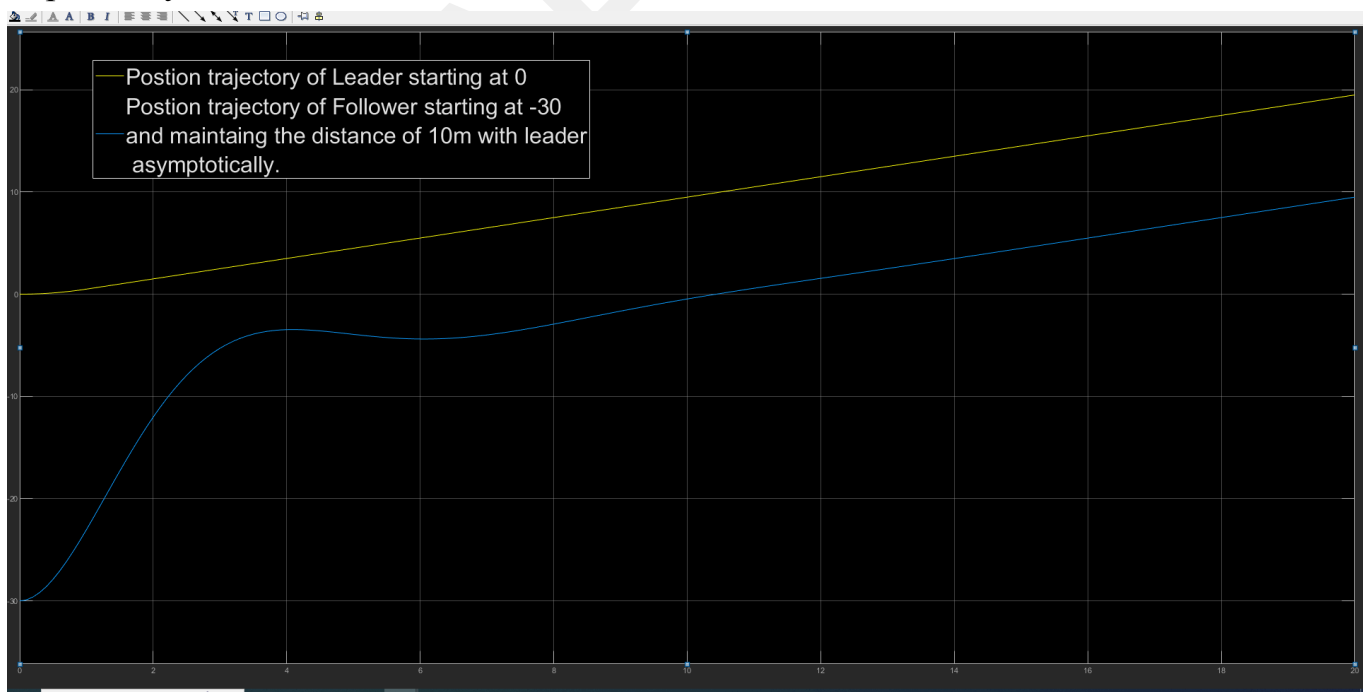


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The subsystems, Leader and Follower 1 have the dynamics as discussed in point 1 and 2 respectively.



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4. Similarly, if one more agent adds to the network as follows (in 1 Dimension):



The dynamics are as follows:

Leader	$\dot{x}_1 = u_a$ $\dot{x}_2 = x_1$
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Follower 1	$\dot{y}_1 = u_{f1}$ $\dot{y}_2 = y_1$
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Case1: To obtain consensus with Leader

$$u_{f1} = -K_1(y_1 - x_1) - K_2(y_2 - x_2).$$

Case2: To maintain a distance of (dist) m with Leader

$$u_{f1} = -K_1(y_1 - x_1) - K_2(dist + y_2 - x_2).$$

Follower 2	$\dot{z}_1 = u_{f2}$ $\dot{z}_2 = z_1$
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Case1: To obtain consensus with Follower 1

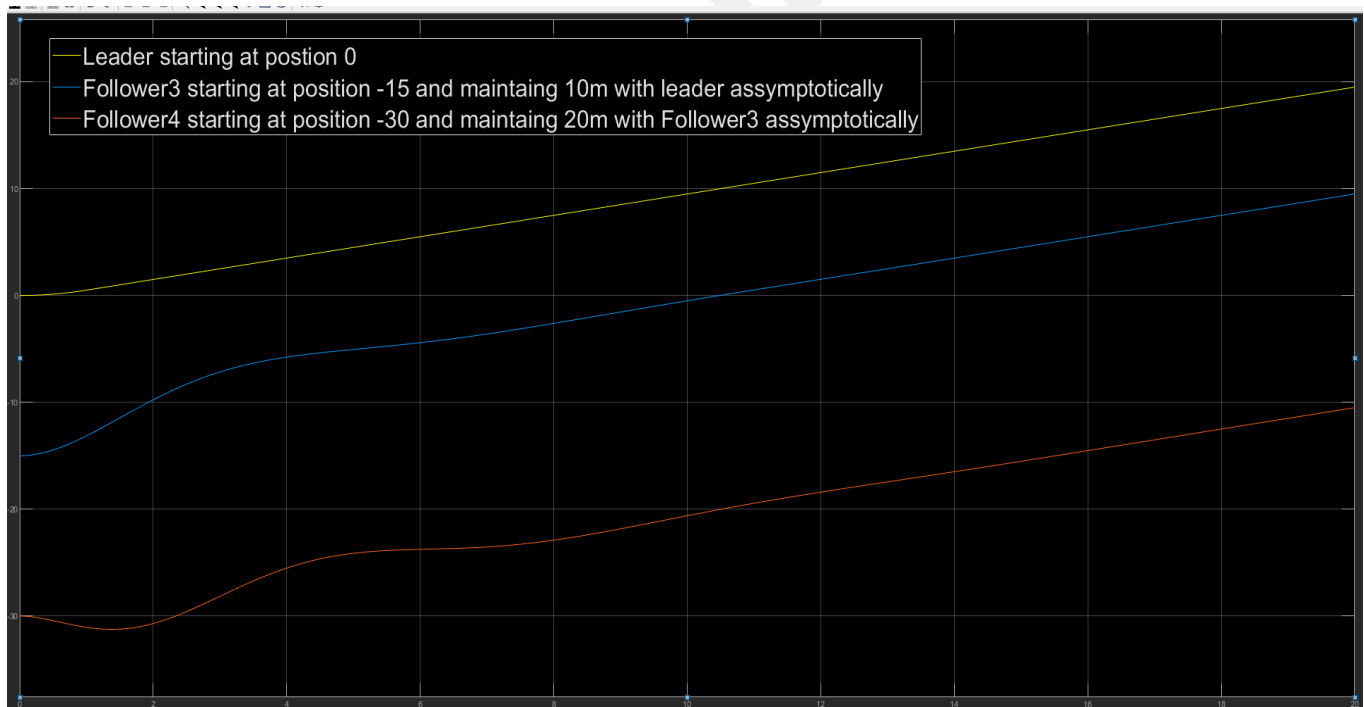
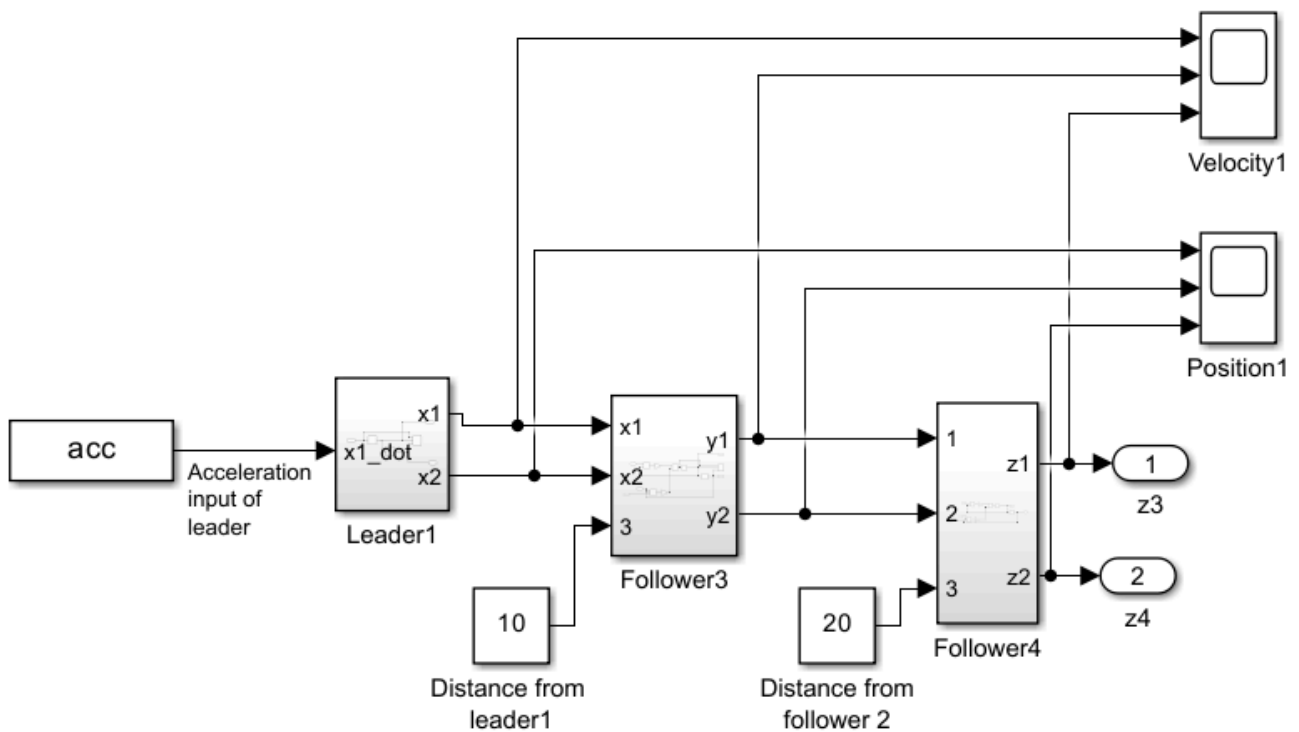
$$u_{f2} = -K_1(z_1 - y_1) - K_2(z_2 - y_2).$$

Case2: To maintain a distance of (dist) m with Follower 2

$$u_{f2} = -K_1(z_1 - y_1) - K_2(dist + z_2 - y_2).$$

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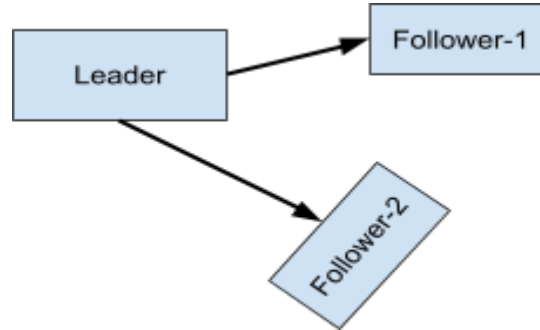
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5. In case, if one more agent adds to the network as follows (in 2 Dimensions):



The dynamics are as follows (now all the states and inputs $\in R^2$ and $K_1, K_2 \in R^{2 \times 2}$):

Leader	$\dot{X}_1 = U_a$ $\dot{X}_2 = X_1$
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Follower 1	$\dot{Y}_1 = U_{f1}$ $\dot{Y}_2 = Y_1$
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Case1: To obtain consensus with Leader

$$U_{f1} = -K_1(Y_1 - X_1) - K_2(Y_2 - X_2).$$

Case2: To maintain a distance of (Dist) m with Leader

$$U_{f1} = -K_1(Y_1 - X_1) - K_2(Dist + Y_2 - X_2).$$

Follower 2	$\dot{Z}_1 = U_{f2}$ $\dot{Z}_2 = Z_1$
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Case1: To obtain consensus with Follower 1

$$U_{f2} = -K_1(Z_1 - X_1) - K_2(Z_2 - X_2).$$

Case2: To maintain a distance of (dist) m with Follower 2

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$$U_{f2} = -K_1(Z_1 - X_1) - K_2(Dist + Z_2 - X_2).$$

Note:

- If the number of Agents increases, it becomes difficult to design control law to every agent individually. So the concept of **Graph theory** is designed where a simultaneous control law is designed for all the agents.
- The Linear dynamics used above are called **Double Integrator Dynamics**.